

EXPLORER

1.java

×

▼ JAVA_...

 1.java

```

1  class Book {
2      private int bookId;
3      private String title;
4      private String author;
5      private double price;
6      private boolean issued;
7
8      public Book(int bookId, String title, String author, double price) {
9          this.bookId = bookId;
10         this.title = title;
11         this.author = author;
12         this.price = price;
13     }
14
15     public int getBookId() { return bookId; }
16     public String getTitle() { return title; }
17     public double getPrice() { return price; }
18
19     public void issueBook() {
20         if (issued) System.out.println(title + " already issued.");
21         else { issued = true; System.out.println(title + " issued."); }
22     }
23
24     public void returnBook() {
25         if (!issued) System.out.println(title + " not issued.");
26         else { issued = false; System.out.println(title + " returned."); }
27     }
28 }

```

 powershell
 







```
PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 1.java
```

- `>> java Sol1`

101 - Java Basics by J. Gosling Rs.350.0 [Available]

102 - Data Structures by N. Wirth Rs.420.0 [Available]

103 - Clean Code by R. Martin Rs.500.0 [Available]

Data Structures issued.

Data Structures already issued.

Data Structures returned.

Found:

103 - Clean Code by R. Martin Rs.500.0 [Available]

```
PS C:\Users\ABDUL AZEEZ\Downloads\Java Coding Challenge>
```

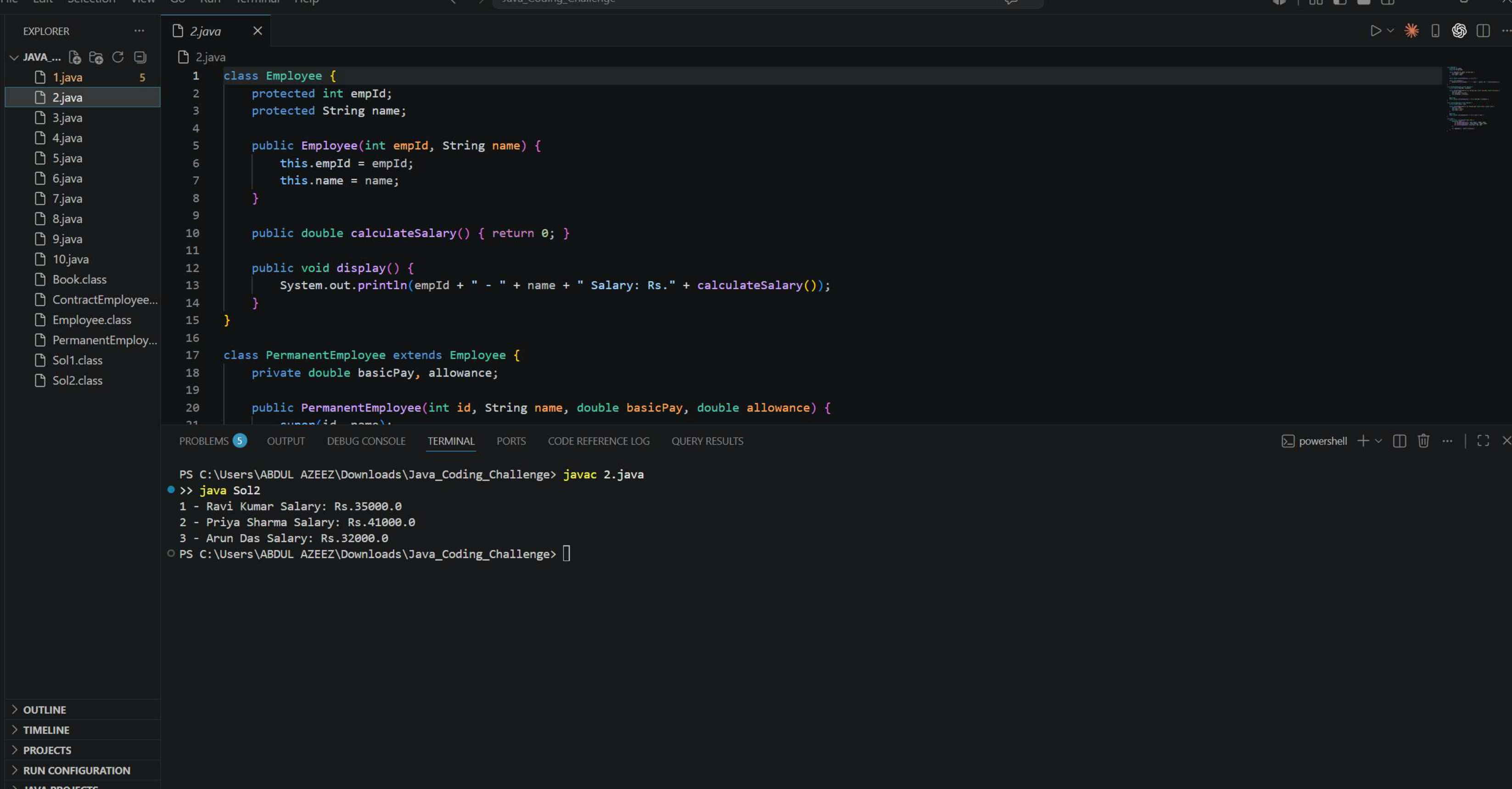
> OUTLINE

> TIMELINE

> PROJECTS

> RUN CONFIGURATION

> JAVA PROJECTS



EXPLORER

JAVA_...

1.java

2.java

3.java

4.java

5.java

6.java

7.java

8.java

9.java

10.java

Book.class

ContractEmployee...

Employee.class

PermanentEmploy...

Sol1.class

Sol2.class

OUTLINE

TIMELINE

PROJECTS

RUN CONFIGURATION

JAVA PROJECTS

2.java

2.java

1 class Employee {

2 protected int empId;

3 protected String name;

4

5 public Employee(int empId, String name) {

6 this.empId = empId;

7 this.name = name;

8 }

9

10 public double calculateSalary() { return 0; }

11

12 public void display() {

13 System.out.println(empId + " - " + name + " Salary: Rs." + calculateSalary());

14 }

15 }

16

17 class PermanentEmployee extends Employee {

18 private double basicPay, allowance;

19

20 public PermanentEmployee(int id, String name, double basicPay, double allowance) {

21 super(id, name);

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 2.java

>> java Sol2

1 - Ravi Kumar Salary: Rs.35000.0

2 - Priya Sharma Salary: Rs.41000.0

3 - Arun Das Salary: Rs.32000.0

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>

3 ×

4.java 3

Vehicle.clas

▶ JAVA PROJECTS

```

1 class BankAccount {
2     private String accNo;
3     private String holder;
4     private double balance;
5
6     public BankAccount(String accNo, String holder, double balance) {
7         this.accNo = accNo;
8         this.holder = holder;
9         this.balance = balance;
10    }
11
12    public void deposit(double amount) {
13        if (amount <= 0) { System.out.println(x: "Invalid deposit."); return; }
14        balance += amount;
15        System.out.println("Deposited Rs." + amount + " New Balance: Rs." + balance);
16    }
17
18    public void withdraw(double amount) {
19        if (amount <= 0) { System.out.println(x: "Invalid withdrawal."); return; }
20        if (amount > balance) { System.out.println(x: "Insufficient balance."); return; }
21        balance -= amount;
22    }
23 }

```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS CODE REFERENCE LOG QUERY RESULTS

```
PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 4.java
>> java Sol4
Karthik R (SB1001) Balance: Rs.10000.0
Deposited Rs.5000.0 New Balance: Rs.15000.0
Withdrew Rs.3000.0 New Balance: Rs.12000.0
Insufficient balance.
Invalid deposit.
Karthik R (SB1001) Balance: Rs.12000.0
PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>
```

EXPLORER

JAVA_CODING_CHALLENGE

1.java5

2.java5

3.java1

4.java3

5.java1

6.java

7.java

8.java

9.java

10.java

Admission.class

BankAccount.class

Bike.class

Book.class

Bus.class

Car.class

ContractEmployee...

Employee.class

PermanentEmploy...

Sol1.class

Sol2.class

Sol3.class

Sol4.class

Sol5.class

Vehicle.class

5.java

Language Support for Java(TM) by Red Hat

Admission

```
1 class Admission {
2     public double calculateFee(String course) {
3         System.out.println("UG Fee for " + course + ": Rs.50000");
4         return 50000;
5     }
6
7     public double calculateFee(String course, String specialization) {
8         System.out.println("PG Fee for " + course + " (" + specialization + "): Rs.80000");
9         return 80000;
10    }
11
12    public double calculateFee(String course, double percentageScored) {
13        double fee = 50000;
14        if (percentageScored >= 90) fee = fee * 0.5;
15        else if (percentageScored >= 75) fee = fee * 0.75;
16        System.out.println("Scholarship Fee for " + course + " (" + percentageScored
17            + "%): Rs." + fee);
18        return fee;
19    }
20 }
```

PROBLEMS15

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

CODE REFERENCE LOG

QUERY RESULTS

powershell

```
PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 5.java
>> java Sol5
UG Fee for B.Tech CSE: Rs.50000
PG Fee for M.Tech (AI & Data Science): Rs.80000
Scholarship Fee for B.Tech CSE (92.0%): Rs.25000.0
Scholarship Fee for B.Tech CSE (78.0%): Rs.37500.0
PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>
```

> OUTLINE

> TIMELINE

> PROJECTS

EXPLORER

6.java 6 X

6.java > Language Support for Java(TM) by Red Hat > Shape

1.java 5

2.java 5

3.java 1

4.java 3

5.java 1

6.java 6

7.java

8.java

9.java

10.java

Admission.class

BankAccount.class

Bike.class

Book.class

Bus.class

Car.class

Circle.class

ContractEmployee...

Employee.class

PermanentEmploy...

Rectangle.class

Shape.class

Sol1.class

Sol2.class

Sol3.class

Sol4.class

Sol5.class

Sol6.class

Triangle.class

Vehicle.class

> OUTLINE

> TIMELINE

> PROJECTS

> RUN CONFIGURATION

> JAVA PROJECTS

1 abstract class Shape {

2 public abstract double area();

3 public void display() {

4 System.out.println(getClass().getSimpleName() + " Area: " + area());

5 }

6 }

7

8 class Circle extends Shape {

9 private double radius;

10 public Circle(double radius) { this.radius = radius; }

11

12 @Override

13 public double area() { return Math.PI * radius * radius; }

14 }

15

16 class Rectangle extends Shape {

17 private double length, width;

18 public Rectangle(double length, double width) {

19 this.length = length;

20 this.width = width;

21 }

PROBLEMS 21

OUTPUT

DEBUG CONSOLE

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PORTS

CODE REFERENCE LOG

QUERY RESULTS

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 6.java

>> java Sol6

Circle Area: 78.53981633974483

Rectangle Area: 24.0

Triangle Area: 12.0

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>

EXPLORER

7.java

5

X

7.java > Language Support for Java(TM) by Red Hat > MedicalRecord

1

interface MedicalRecord {

2

void addRecord();

3

void displayRecord();

4

}

5

6

class Patient implements MedicalRecord {

7

private String name;

8

private String disease;

9

10

public Patient(String name, String disease) {

11

this.name = name;

12

this.disease = disease;

13

}

14

15

@Override

16

public void addRecord() {

17

System.out.println("Patient record added for " + name);

18

}

19

20

@Override

21

public void displayRecord() {

PROBLEMS

26

OUTPUT

DEBUG CONSOLE

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PORTS

CODE REFERENCE LOG

QUERY RESULTS

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 7.java

• >> java Sol7

Patient record added for Suresh Kumar

Patient: Suresh Kumar | Disease: Fever

Doctor record added for Dr. Anitha

Doctor: Dr. Anitha | Specialization: Cardiology

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>

1.java

5

2.java

5

3.java

1

4.java

3

5.java

1

6.java

6

7.java

5

8.java

9.java

10.java

Admission.class

BankAccount.class

Bike.class

Book.class

Bus.class

Car.class

Circle.class

ContractEmployee...

Doctor.class

Employee.class

MedicalRecord.class

Patient.class

PermanentEmploy...

Rectangle.class

Shape.class

Sol1.class

Sol2.class

Sol3.class

Sol4.class

Sol5.class

Sol6.class

OUTLINE

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PROJECTS

RUN CONFIGURATION

JAVA PROJECTS

EXPLORER

9.java 4 X

9.java > Language Support for Java(TM) by Red Hat > Student

Amazon Q: Edit (Ctrl+I)

```
1 class Student {
2     private int roll;
3     private String name;
4     private int[] marks;
5
6     public Student(int roll, String name, int[] marks) {
7         this.roll = roll;
8         this.name = name;
9         this.marks = marks;
10    }
11
12    public int getTotal() {
13        int total = 0;
14        for (int m : marks) total += m;
15        return total;
16    }
17
18    public double getAverage() { return getTotal() / 3.0; }
19
20    public String getGrade() {
21        double avg = getAverage();
```

1.java 5

2.java 5

3.java 1

4.java 3

5.java 1

6.java 6

7.java 5

8.java 4

9.java 4

10.java

Admission.class

BankAccount.class

Bike.class

Book.class

Bus.class

Car.class

Circle.class

ContractEmployee...

Doctor.class

Employee.class

MedicalRecord.class

Order.class

Patient.class

PermanentEmploy...

Product.class

Rectangle.class

Shape.class

Sol1.class

Sol2.class

Sol3.class

Sol4.class

OUTLINE

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JAVA PROJECTS

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CODE REFERENCE LOG

QUERY RESULTS

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 9.java

>> java Sol9

1 - Aravind Total: 263 Avg: 87.66666666666667 Grade: A

2 - Divya Total: 284 Avg: 94.66666666666667 Grade: A+

3 - Karthik Total: 180 Avg: 60.0 Grade: B

Class Topper: Divya

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>

EXPLORER

...

10.java 1 X

10.java > Language Support for Java(TM) by Red Hat > Device

1.java 5

2.java 5

3.java 1

4.java 3

5.java 1

6.java 6

7.java 5

8.java 4

9.java 4

10.java 1

Admission.class

AirConditioner.class

BankAccount.class

Bike.class

Book.class

Bus.class

Car.class

Circle.class

ContractEmployee...

Device.class

Doctor.class

Employee.class

Fan.class

Light.class

MedicalRecord.class

Order.class

Patient.class

PermanentEmploy...

Product.class

Rectangle.class

Shape.class

1 class Device {

2 protected String name;

3 protected boolean on;

4

5 public Device(String name) { this.name = name; }

6

7 public void turnOn() { on = true; System.out.println(name + " turned ON."); }

8 public void turnOff() { on = false; System.out.println(name + " turned OFF."); }

9

10 public double powerConsumption() { return 0; }

11

12 public void display() {

13 System.out.println(name + " [" + (on ? "ON" : "OFF") + "] Power: "

14 + powerConsumption() + "W");

15 }

16 }

17

18 class Light extends Device {

19 public Light(String name) { super(name); }

20

21 }

PROBLEMS 35

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

CODE REFERENCE LOG

QUERY RESULTS

powershell + v

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge> javac 10.java

>> java Sol10

Hall Light turned ON.

Bedroom Fan turned ON.

Living Room AC turned ON.

Hall Light [ON] Power: 10.0W

Bedroom Fan [ON] Power: 60.0W

Living Room AC [ON] Power: 1500.0W

Living Room AC turned OFF.

Living Room AC [OFF] Power: 0.0W

PS C:\Users\ABDUL AZEEZ\Downloads\Java_Coding_Challenge>

OUTLINE